System Capacity & Care Load Analytics: Internship Project Report

Author: Reena Jesupa S | **Program:** Unified Mentor Machine Learning Internship | **Code:** github.com/Reena-1-1/system_capacity_analytics | Live dashboard: https://systemcapacityanalytics-online.streamlit.app/

Executive Summary

This project analyses 720 daily care-system records spanning 12 Jan 2023 to 21 Dec 2025 (355 calendar days have no report), measures pressure across CBP custody and HHS care, and forecasts total system load 30 days ahead in an interactive Streamlit dashboard.

Three findings stand out:

1. Total system load peaked at 11,762 children and averaged 6,233 per day.
2. HHS care load fell sharply in early 2025 and has stayed near 2,500 (latest total: 2,502).
3. Average net HHS intake was -44.7 per day, so discharges exceeded transfers over the period.

A naive baseline and an ARIMA(1,1,1) model were compared on a held-out final 20% of the data. The naive baseline was slightly better (MAE 229.36 vs 245.12), so the 30-day forecast stays almost flat at about 2,495 to 2,499 children. The data records counts, not bed capacity, and 355 of 1,075 calendar days have no report, so the results show pressure and trend, not proven overcrowding.

Abstract

This project builds an end-to-end time-series analytics pipeline for a daily CBP and HHS care dataset covering January 2023 to December 2025. The data was cleaned and validated, explored visually, and turned into KPIs such as total system load, net HHS intake, growth rate and rolling averages. A naive baseline and an ARIMA(1,1,1) model were trained on the first 80% of days and tested on the last 20%, then ARIMA was refit on all data to produce a 30-day forecast. The results are delivered in a Streamlit dashboard with date filters, KPI cards and forecast charts. The naive baseline (MAE 229.36) slightly outperformed ARIMA(1,1,1) (MAE 245.12) on the test period.

1. Introduction

Care-system managers need to know how many children are in each stage, when pressure builds, and what the load may look like in coming weeks. Daily counts exist, but raw tables do not show trends or early warning signs.

Objectives

- Clean and validate the raw daily records.
- Explore trends and relationships between CBP and HHS measures.
- Engineer KPIs that describe system pressure.
- Forecast total system load and test the forecast on unseen data.
- Present the results in a deployable dashboard.

2. Dataset Description

The dataset is the HHS Unaccompanied Alien Children Program daily file, covering 12 Jan 2023 to 21 Dec 2025 with 720 rows (one per reported day; 355 calendar days have no report) and six columns.

Column	Meaning
Date	Reporting date
Children apprehended and placed in CBP custody	Daily intake into CBP
Children in CBP custody	Children held by CBP at reporting time
Children transferred out of CBP custody	Daily transfers from CBP to HHS
Children in HHS Care	Children in HHS care at reporting time
Children discharged from HHS Care	Daily discharges from HHS

3. Data Cleaning and Validation

The raw file needed cleaning before analysis. A footnote asterisk and hidden characters in the headers caused a `KeyError`, and a missing comma in a column list had merged “Date” with the first column name.

1. **Column names:** removed the BOM character, asterisks and extra spaces, then set exact names by position.
2. **Dates:** parsed with `pd.to_datetime`, dropped 450 blank rows (no date and no counts) and sorted ascending.
3. **Numbers:** removed thousands separators and converted the five count columns to numeric.
4. **Validation flags:** marked days where transfers exceeded CBP custody or discharges exceeded HHS care.
5. **Gaps:** found 355 missing calendar dates out of 1,075 and filled them by linear interpolation for forecasting only, never with zero.

Check	Result
Rows in the raw file	1,170
Blank rows dropped (no date and no counts)	450
Rows after cleaning	720
Expected calendar days (12 Jan 2023 to 21 Dec 2025)	1,075
Missing calendar dates	355 (33%)
Duplicate dates	0
Missing values in any column	0
Transfers exceed CBP custody (flagged days)	86
Discharges exceed HHS care (flagged days)	0

Flagged rows are quality indicators, not proven errors, because daily flows and end-of-day counts may follow different reporting conventions.

4. Exploratory Data Analysis

HHS care load is far larger than CBP custody throughout the period. It rose to roughly 11,000 around late 2023 and early 2024, fell to about 6,000 through 2024, then dropped sharply in early 2025 to around 2,000 to 2,500. CBP custody stays small by comparison and follows the same downturn.

The daily care load, intake vs discharges, monthly average and correlation charts are in `notebooks/EDA.ipynb` and in the live dashboard.

Intake and discharges both fall to near zero from early 2025, matching the drop in care load.

5. Feature Engineering and KPIs

Feature	Definition
Total System Load	Children in CBP custody + children in HHS care
Net HHS Intake	Transfers out of CBP − discharges from HHS
Daily System Load Change	Day-over-day difference in total load
Growth Rate (%)	Day-over-day percentage change in total load
7-Day and 14-Day Average	Rolling mean of total load
Sustained HHS Pressure	Positive net intake on at least 5 of the last 7 days

KPI	Value
Total children under care (latest)	2,502
Average daily system load	6,233
Maximum recorded system load	11,762
Net intake pressure (average net HHS intake)	-44.7 per day
Care load volatility index	0.020 (7-record standard deviation divided by 7-record average)
Backlog accumulation rate	-1.3 (latest 7-record average of net HHS intake)
Discharge offset ratio	1.35 (total discharges divided by total transfers)
Positive / negative net intake days	238 positive / 475 negative (7 days at zero)

Net HHS Intake measures pressure on HHS, not the net change of the whole system, because transfers move children inside it.

Strain and relief periods

A day is classed as strain when its 7-day average load is in the top 25% of the full period, and as relief when it is in the bottom 25%. Runs of 14 or more consecutive strain records (days with a report) are listed as prolonged strain windows. Because some calendar days have no report, the calendar span of each window is longer than its record count.

Window	Start	End	Records	Calendar span (days)
1	1 May 2023	23 May 2023	17	23
2	7 Aug 2023	22 Jan 2024	111	169
3	25 Jan 2024	27 Mar 2024	44	63

Windows 2 and 3 are only 2 days apart, so the system was under sustained strain for about 7.5 months (7 Aug 2023 to 27 Mar 2024). No prolonged strain window appears after March 2024.

Early vs late comparison

The timeline was split at its midpoint. In the early half, average total load was 8,615 (maximum 11,762) and net HHS intake averaged -88.3 per day. In the late half, average load was 3,864 (maximum 7,016) and net intake averaged -1.4 per day. Load fell by more than half, and net intake moved from heavy discharging to near balance. Volatility stayed low in both halves (average under 0.05, maximum about 0.1).

Weekly trend

The weekly average smooths daily noise and follows the same overall pattern as the daily series: a build-up through 2023, a peak around the turn of 2023 to 2024, and a sharp drop in early 2025.

6. Forecasting

The target is Total System Load. The series was set to a regular daily index, with any gaps interpolated, and split in time order: the first 80% of days for training and the last 20% for testing. A random split would leak future information into training.

- **Baseline:** repeat the last training value for every test day.
- **ARIMA(1,1,1):** a statistical model of past values and past errors, fitted on the training set.
- **Metrics:** MAE is the average absolute error in children. RMSE penalises large errors more. Accuracy is not used because this is a regression task.

Model	MAE	RMSE
Naive baseline	229.36	284.48
ARIMA(1,1,1)	245.12	301.98

The naive baseline performed slightly better: its MAE was 229.36 against 245.12 for ARIMA (about 7% higher), and its RMSE was 284.48 against 301.98. The test period is the final 20% of days, a stable low-load stretch after the early 2025 drop, where repeating the last value is hard to beat. ARIMA(1,1,1) therefore added no value on this split. The next step would be SARIMA or a lag-feature model. The 30-day forecast is almost flat, from 2,499 on its first day to 2,495 on its last, close to the latest observed load of 2,502.

After evaluation, the model was refit on all data to produce a 30-day forecast, saved in `forecast_30_days.csv`. The actual-vs-predicted and 30-day forecast charts are in the notebook and the live dashboard.

7. Streamlit Dashboard

The dashboard (`app.py`) loads the raw CSV, applies the same cleaning, and offers a date-range filter, time granularity (daily, weekly, monthly), metric toggles, five KPI cards, a CBP vs HHS comparison, net intake and backlog trends, a strain and relief table, and a 30-day ARIMA forecast. It is live at https://systemcapacityanalytics-online.streamlit.app/ and the code is at https://github.com/Reena-1-1/system_capacity_analytics.

8. Key Findings and Recommendations

1. Load peaked at 11,762 and fell sharply in early 2025, so current pressure is far below the 2023 to 2024 peak (average load fell from 8,615 in the early half to 3,864 in the late half).
2. Discharges exceeded transfers on average (net HHS intake -44.7 per day; 475 negative vs 238 positive days).
3. A naive baseline beat ARIMA(1,1,1) on the held-out period (MAE 229.36 vs 245.12), so the forecast is a stable-level estimate of about 2,495 to 2,499 children, not a demonstrated improvement over simple persistence.
4. **Recommendation:** track the 7-day average and the sustained-pressure flag as early warnings.
5. **Recommendation:** collect bed-capacity data and complete daily reporting, so utilisation can be measured and gaps do not need interpolation.

9. Limitations

- The dataset has no bed capacity, so the project shows pressure, not overcrowding.
- 355 of 1,075 calendar days (33%) have no report. Gaps were filled by interpolation for forecasting only. Rolling averages, the volatility index, backlog rate and strain windows are computed over the last 7 or 14 reported records, not calendar days.
- On 86 reported days, transfers exceeded the recorded CBP custody count, which may reflect differing reporting conventions.
- ARIMA(1,1,1) was a starting configuration, was not tuned, and did not beat the naive baseline.
- Forecasts are statistical estimates, not official projections.

10. Conclusion

The project turned raw daily records into a cleaned dataset, a set of pressure KPIs, a tested forecast and a deployed dashboard. The main lesson is that a simple model must beat a simple baseline before it is trusted. Future work includes SARIMA, lag-feature machine learning models, alert thresholds and automated data refresh.

References

- HHS Unaccompanied Alien Children Program, daily data file
- Python libraries: pandas, NumPy, Plotly, Matplotlib, Seaborn, statsmodels, scikit-learn, Streamlit